

Ecommerce Product Discovery & Conversion

Analytics Platform

Industry-style analytics report using Python, DuckDB SQL, and Power BI to analyze product discovery, funnel drop-offs, product ranking opportunities, category/brand performance, and user segmentation from the REES46 ecommerce behavior dataset.

Prepared by: **Piyush Kumar**

Portfolio Project - Data Analyst / Ecommerce Analytics

\$47.38M Combined Revenue Oct-Nov 2019	6.55M Sessions combined	3.10M Users combined	268K+ Products analyzed
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Report note: This is an independent portfolio project using public ecommerce behavior data. It is inspired by product discovery analytics and is not an official Constructor.io project or dataset.

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The report is written for a business and analytics audience. It highlights the decision-making value of the dashboard rather than only explaining code.

1. Executive Summary

DiscoverIQ is an ecommerce product discovery analytics project built to understand how users move from product view to add-to-cart to purchase. The project analyzes October and November 2019 behavior from the public REES46 ecommerce dataset and turns raw event data into SQL outputs and a Power BI decision dashboard.

Core business finding: November generated more traffic and stronger product interest, but revenue declined because purchase completion weakened sharply.

\$27.02M October Revenue baseline month	\$20.36M November Revenue -24.65%	13.68% October Abandonment cart sessions	67.73% November Abandonment +54.05 pts
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Metric	October	November	Change
Sessions	3.12M	3.43M	+9.70%
Users	1.50M	1.60M	+6.89%
Revenue	\$27.02M	\$20.36M	-24.65%
Conversion Rate	2.82%	2.05%	-0.77 pts
Cart Abandonment	13.68%	67.73%	+54.05 pts
Revenue per Session	\$8.65	\$5.94	-\$2.71

The dashboard shows that the November problem was not traffic generation. Users were browsing and adding items to cart, but they were not converting at the same rate. This creates a clear business priority: reduce cart abandonment, improve purchase confidence, and optimize the product discovery-to-purchase journey.

2. Dataset and Scale

The project uses public ecommerce behavior data from the REES46 Kaggle dataset. The raw data contains event-level product interactions such as views, carts, and purchases for a large multi-category online store.

Area	Details
Dataset	REES46 Ecommerce Behavior Dataset from Kaggle
Period Used	October and November 2019
Raw Scale	42M+ rows in October and 67M+ rows in November
Working Data	5M stratified rows per month
Final Combined Sessions	6,548,069
Final Combined Users	3,097,396
Products Analyzed	268,447
Brands Analyzed	726
Combined Revenue	\$47.38M

Large-data learning: the raw files were too large for a normal free notebook workflow. This required stratified sampling, memory optimization, and aggregation into BI-ready tables.

Stratified sampling was done by event type so the sample preserved the original view/cart/purchase distribution. This was essential because the project depends on reliable funnel analysis.

Data Challenge	Solution Applied	Why It Mattered
Raw files too large	5M-row stratified sample per month	Prevented notebook crashes while preserving behavior
High RAM usage	Downcasting and category types	Reduced memory footprint
Power BI performance	Summary BI exports	Dashboard stayed responsive
Missing category/brand	Filled with unknown	Kept behavior without hiding data quality issues

3. Data Engineering Methodology

The project workflow moved from raw event data to optimized analytical tables and then to Power BI-ready exports. This is similar to real BI workflows where dashboards are not built directly on raw event logs.

Stage	Method	Output
Data Loading	Kaggle Notebook and Pandas	Loaded October and November event data
Sampling	Stratified by event_type	5M rows per month
Cleaning	Null handling and zero-price filtering	Valid analytical data
Feature Engineering	Date, week, day, category levels, price tiers	New analysis dimensions
Optimization	Downcasting and category conversion	Memory reduced from 2.94 GB to 0.84 GB
Aggregation	Session/product/user/brand/category tables	Analysis-ready data model
SQL Layer	DuckDB SQL	BI-ready output tables
Visualization	Power BI	Interactive dashboard pages

Analytical table design

Table	Grain	Main Use
combined_session_summary	One row per session	Funnel, revenue, daily trends
combined_dim_product	One row per product per month	Product ranking and conversion
combined_dim_user	One row per user per month	Segmentation and personalization
combined_dim_brand	One row per brand per month	Brand revenue and conversion
combined_dim_category	One row per category per month	Category performance

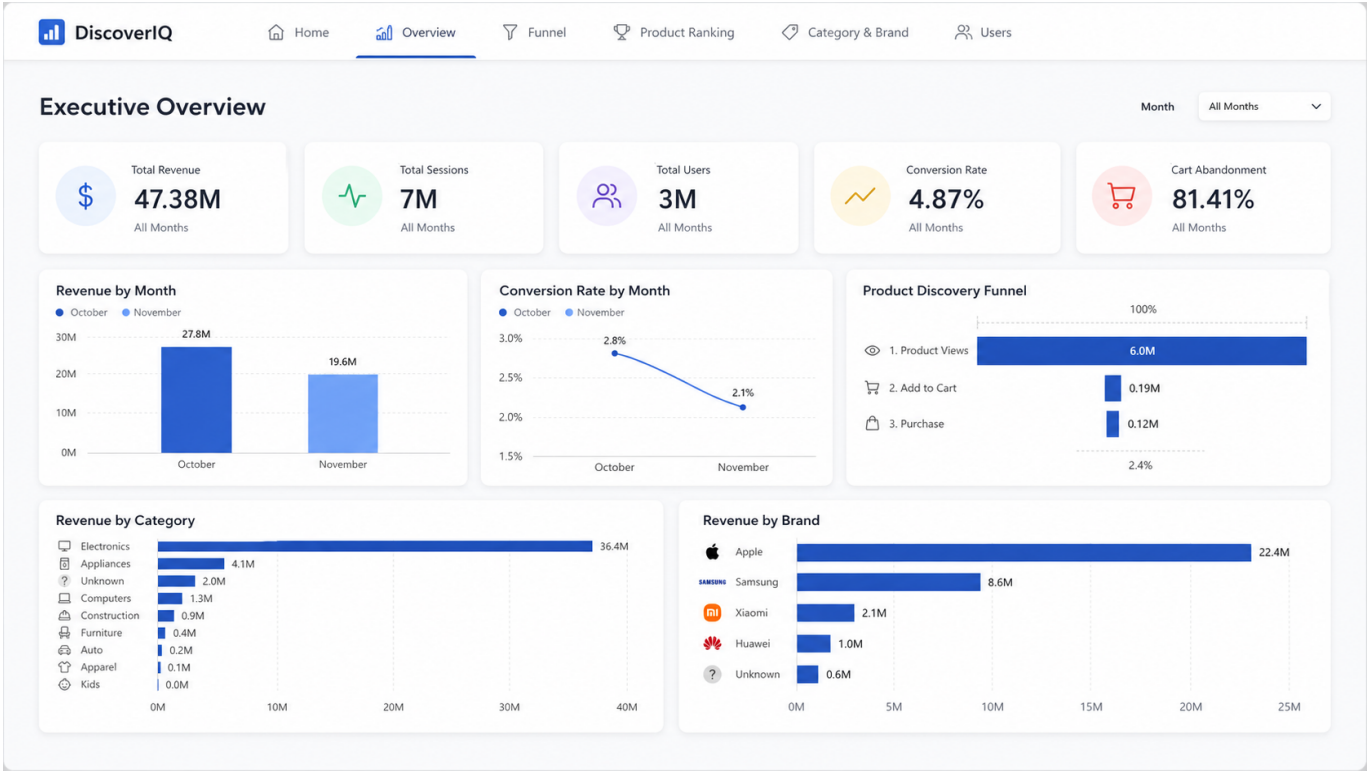
Methodology takeaway: the project demonstrates end-to-end analytics, including large-data handling, memory-aware processing, SQL modeling, and dashboard storytelling.

4. Dashboard Architecture

The Power BI report is structured around business questions. Each page has a clear purpose, controlled scope, and supporting BI export tables.

Dashboard Page	Business Question	Primary Output
Home	What is the project and how should users navigate?	Project entry and summary
Executive Overview	Did performance improve or decline?	Revenue, sessions, conversion, abandonment
Product Discovery Funnel	Where does the journey break?	View-cart-purchase funnel and drop-offs
Product Ranking Analytics	Are the right products getting visibility?	Hidden gems and weak visible products
Category & Brand Performance	Which business areas drive or lose revenue?	Category and brand decline analysis
User Segmentation	Which users create the most value?	New, returning, loyal, and high-value users

Dashboard design preview

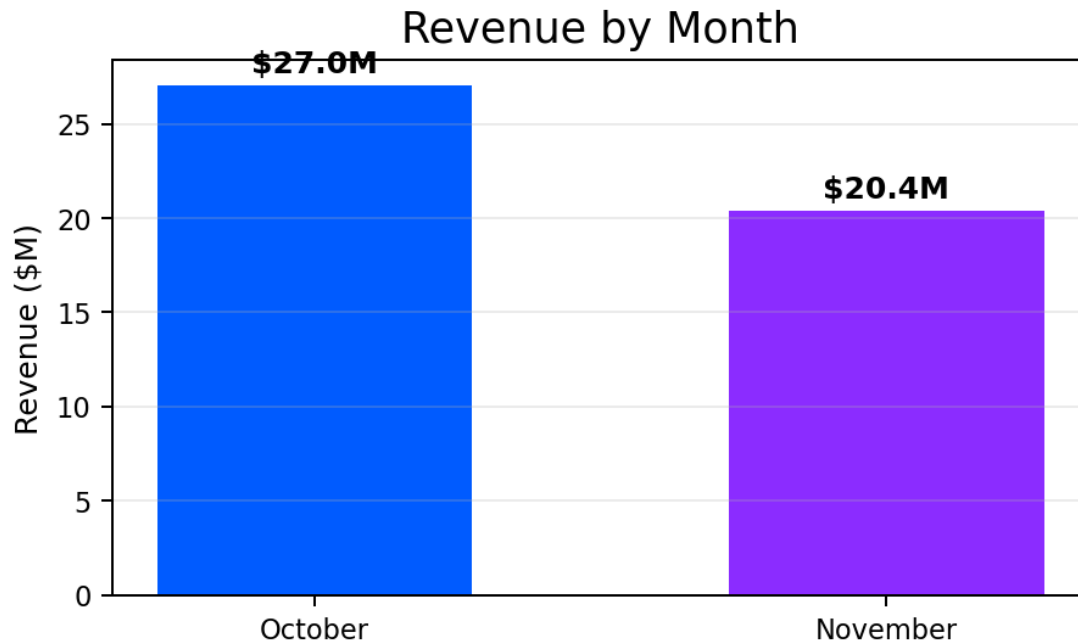


Note: dashboard image is used as a visual design preview. Final numbers should always come from the BI export tables and Power BI model.

5. Business Findings

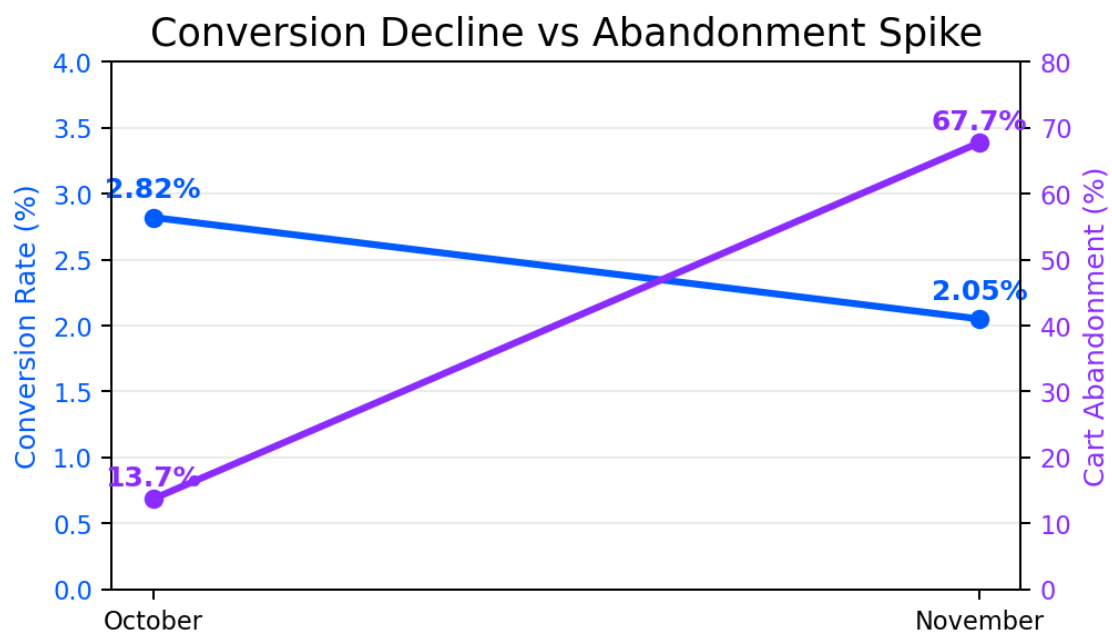
Finding 1: November had more traffic but lower revenue.

Sessions increased by 9.70% and users increased by 6.89%, but revenue declined by 24.65%. This is a classic case where traffic growth creates a false positive unless conversion quality is also analyzed.



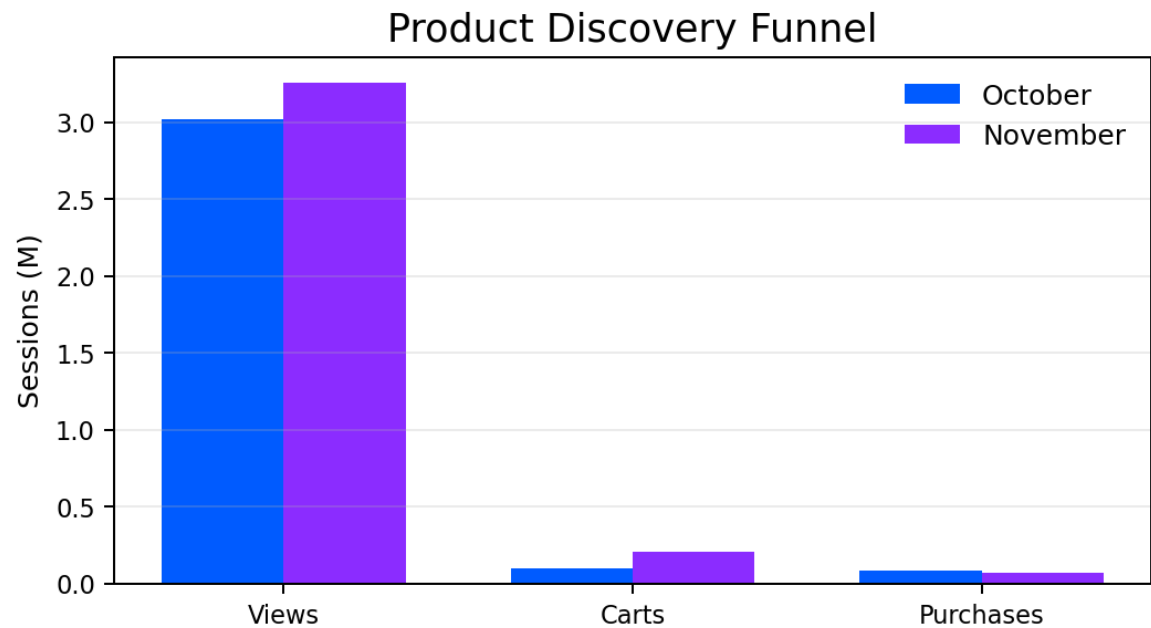
Finding 2: Cart abandonment was the biggest operational problem.

Cart abandonment increased from 13.68% to 67.73%. This indicates that users showed product interest but did not finish the purchase journey.



5. Business Findings - Funnel Deep Dive

The funnel reveals the actual behavior change. November improved view-to-cart rate, but cart-to-purchase conversion collapsed.

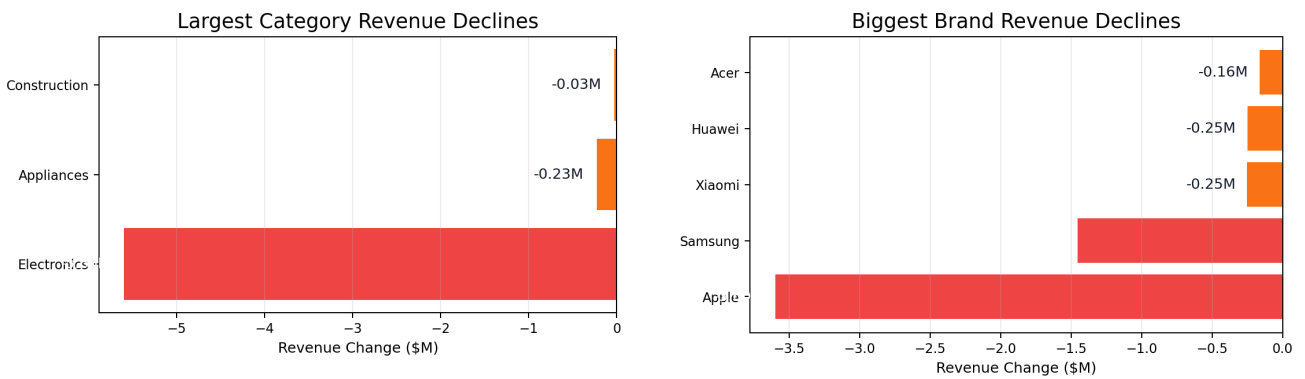


Funnel Metric	October	November	Business Meaning
View-to-Cart Rate	3.27%	6.35%	Product interest improved
Cart-to-Purchase Rate	86.32%	32.27%	Purchase completion collapsed
View-to-Purchase Rate	2.82%	2.05%	Final conversion declined

Interpretation: November users were likely browsing, comparing, and saving items for later. The business should focus on purchase confidence and cart recovery rather than only driving more traffic.

6. Category and Brand Performance

Electronics was the largest revenue category in both months, but it also caused the biggest revenue decline. Apple and Samsung remained the leading brands, yet both dropped significantly in November.



Area	October Revenue	November Revenue	Change
Electronics	\$20.73M	\$15.14M	-\$5.60M
Apple	\$13.00M	\$9.40M	-\$3.60M
Samsung	\$5.49M	\$4.04M	-\$1.45M

Business meaning: focus the first optimization sprint on electronics and top brands, because they have the highest revenue impact. A small conversion improvement in these areas has larger business value than optimizing low-revenue categories first.

7. Product Ranking Analysis

The product ranking page turns raw product metrics into action segments. The goal is to identify which products deserve visibility and which products may be wasting ranking space.

Segment	Definition	Business Action
Winner Product	High visibility and strong conversion	Keep ranking high
Hidden Gem	Lower visibility but strong conversion	Promote higher
High Visibility Low Conversion	High visibility but weak conversion	Investigate or demote
Low Priority Product	Low visibility and weak conversion	Low focus

Hidden gem logic

Rule	Threshold
Views	20 to 200
Conversion Rate	$\geq 3\%$
Purchases	≥ 3

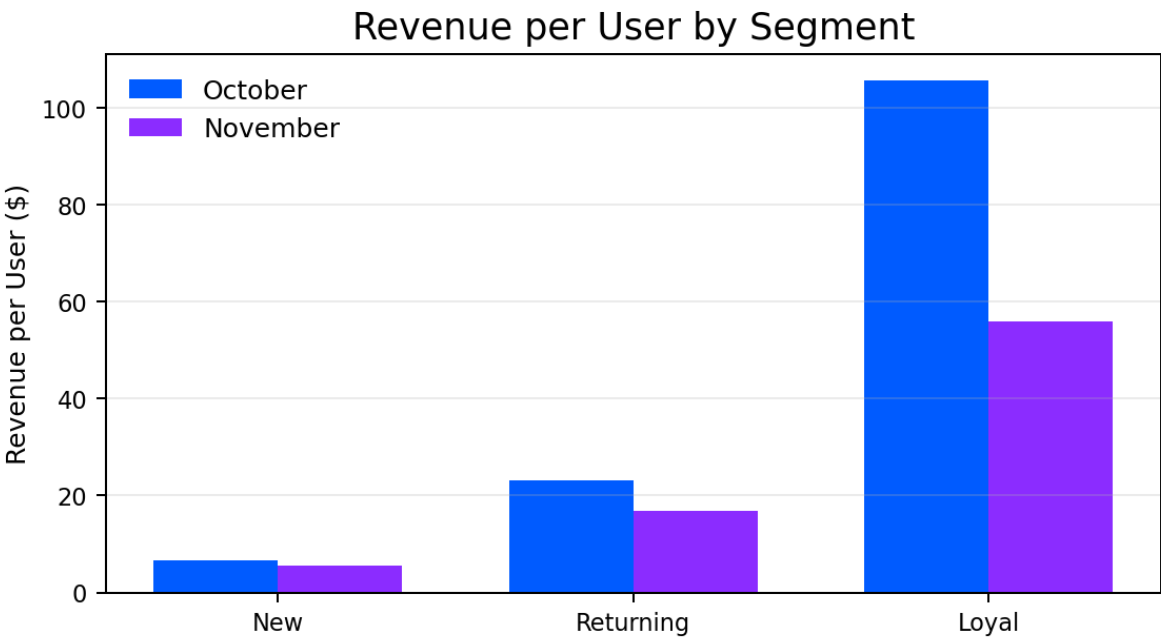
High-visibility low-conversion logic

Rule	Threshold
Views	$\geq 1,000$
Conversion Rate	$< 1\%$
Purchases	< 10

Business meaning: product discovery should not reward visibility alone. Products should be evaluated by how well they convert once users see them.

8. User Segmentation and Personalization

Users were segmented into new, returning, and loyal based on session frequency. Loyal users generated the highest revenue per user, but their spending dropped sharply in November.



Segment	Definition	October Revenue/User	November Revenue/User	Personalization Action
New	1 session	\$6.64	\$5.55	Build trust and guide first purchase
Returning	2-5 sessions	\$23.06	\$16.90	Use behavior-based recommendations
Loyal	6+ sessions	\$105.71	\$55.92	Protect with offers and early access

Insight: loyal users remain the most valuable segment even when revenue per user declines. This group should be prioritized for retention, personalized offers, and cart recovery.

9. Business Recommendations

The recommendations are prioritized by business impact and connection to the dashboard findings.

Priority	Action Area	Reason	Recommended Action
1	Reduce Cart Abandonment	Abandonment jumped to 67.73%	Cart recovery, urgency, checkout optimization
2	Optimize Electronics	Largest revenue driver and largest decline	Improve ranking, offers, comparison, and product content
3	Protect Apple and Samsung	Top brands declined heavily	Brand-specific offers and high-intent retargeting
4	Promote Hidden Gems	Low visibility but strong conversion	Boost in search, browse, and recommendations
5	Investigate Weak Visible Products	High views with weak conversion	Review price, image, content, relevance, and ranking
6	Personalize Loyal Users	Highest revenue per user	Early access, loyalty offers, premium recommendations

90-day execution roadmap

Timeline	Focus	Expected Outcome
0-30 days	Cart abandonment dashboarding and recovery campaigns	Recover lost purchase intent
31-60 days	Product ranking review: hidden gems and weak visible products	Improve product discovery quality
61-90 days	Segment-based personalization for returning and loyal users	Increase repeat purchase value

10. Constructor-Style Product Discovery Alignment

DiscoverIQ is an independent portfolio project, but its business logic aligns with ecommerce product discovery problems: ranking, searchandising, recommendations, browse optimization, personalization, and conversion improvement.

DiscoverIQ Analysis	Product Discovery Use Case
Product view to cart drop-off	Search and browse relevance optimization
Hidden-gem products	Boost underexposed high-converting products
High-view low-conversion products	Searchandising and ranking investigation
Category decline	Category page optimization
Brand conversion gaps	Brand-level merchandising strategy
User segments	Personalization
Co-purchase patterns	Recommendation opportunities
Cart abandonment	Urgency signals and conversion nudges

Positioning statement: DiscoverIQ demonstrates how analytics can turn raw ecommerce behavior into product discovery decisions that improve ranking, conversion, and personalization.

11. Limitations and Future Work

A professional analytics report should clearly explain what the data can and cannot support.

Limitation	Impact	Future Improvement
No search query text	Cannot do true query-level search analytics	Add search logs and query-result data
No recommendation impressions	Cannot measure recommendation CTR or uplift	Track recommendation impressions, clicks, and purchases
Row-level sampling	Can create session artifacts	Use full session-preserving extraction
Session-based abandonment	May miss cross-session purchases	Use cart ID and order lifecycle data
No marketing channels	Cannot attribute traffic source impact	Add UTM and campaign data
No margin, returns, stock	Revenue is not net profit	Add inventory, margin, and return data

Future enhancements

- Build a ranking score using conversion, revenue per view, price tier, and user segment signals.
- Add recommendation logic using co-purchase and category affinity.
- Build a Streamlit version for interactive portfolio demo.
- Add DAX weighted measures for production-grade percentage aggregation.
- Connect to a database pipeline instead of static CSV exports.

Appendix: KPI Definitions

KPI	Formula	Meaning
View-to-Cart Rate	$\text{cart_sessions} / \text{view_sessions} * 100$	Early product interest after product view
Cart-to-Purchase Rate	$\text{purchase_sessions} / \text{cart_sessions} * 100$	Purchase completion after cart
Conversion Rate	$\text{purchase_sessions} / \text{view_sessions} * 100$	Final conversion from product discovery
Cart Abandonment	$(\text{cart_sessions} - \text{purchase_sessions}) / \text{cart_sessions} * 100$	Cart sessions not completed as purchases
Revenue per Session	$\text{total_revenue} / \text{total_sessions}$	Average revenue generated per session
Revenue per User	$\text{total_revenue} / \text{total_users}$	Average value per user
Revenue per View	$\text{total_revenue} / \text{view_sessions}$	Monetization of product visibility

Important Power BI note: percentage fields in the BI exports are stored as percentage values. For example, 2.82 means 2.82%, not 0.0282. Do not multiply these values by 100 again.

End of report